

PETsurfer-BIDS: a BIDS App for Voxel- and Vertex-wise Kinetic Modeling of PET Data

Douglas Greve[1,2], Paul Wighton[1,2], Martin Norgaard[3,4], Granville Matheson[5], Melanie Ganz[3,6], Cyril Pernet[6], Anthony Galassi[4], Gitte Knudsen[6,7], Robert Innis[4]

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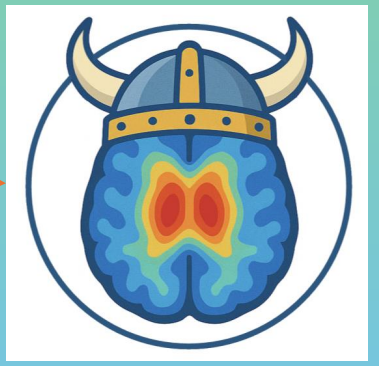






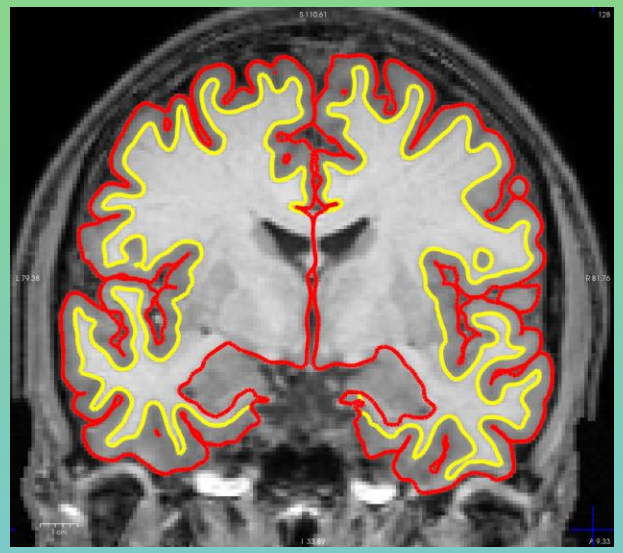
PETsurfer BIDS App Overview

- PETsurfer developed in 2014 (FreeSurfer)
- Ingests PETprep and Bloodstream BIDS derivatives
- Creates BIDS derivatives
- Offers voxel-wise KM of PET data
  - Volume
  - “Pure” Surface
  - Easy to use
- Benefits of Surface-based Analysis
- Individual Analysis in PETsurfer
- Group Analysis in PETsurfer

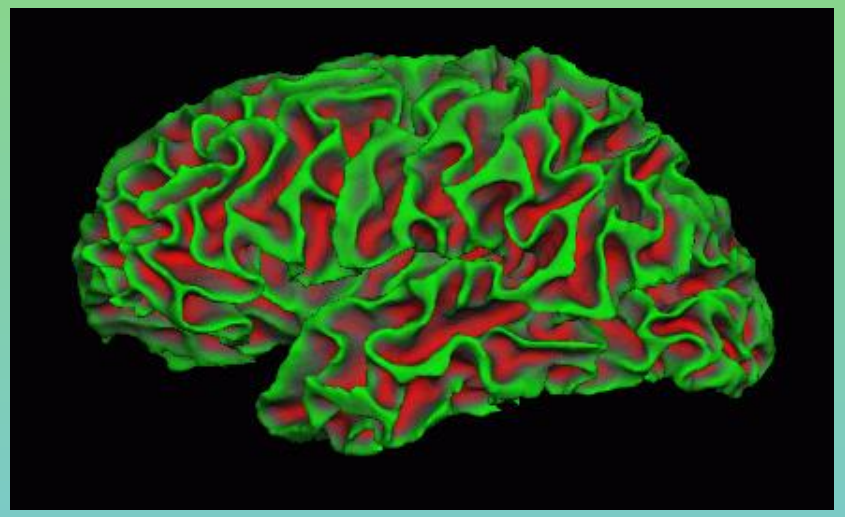




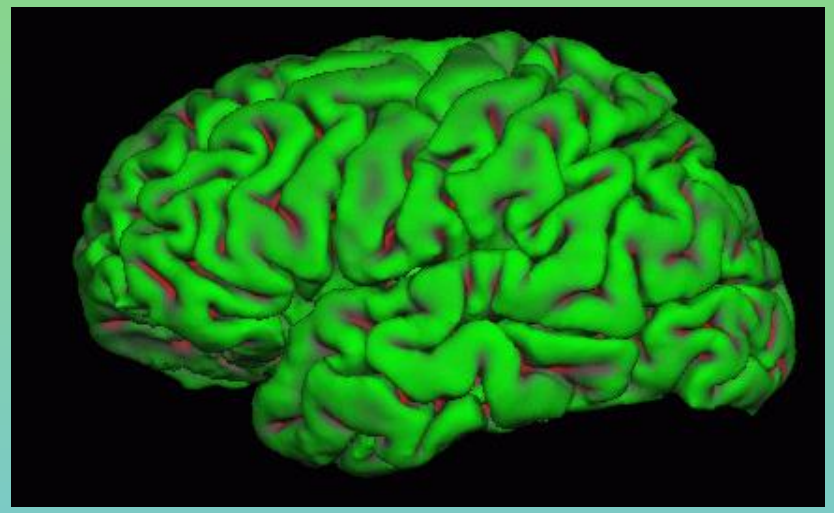
Cortex is a 2D Surface Folded in 3D space



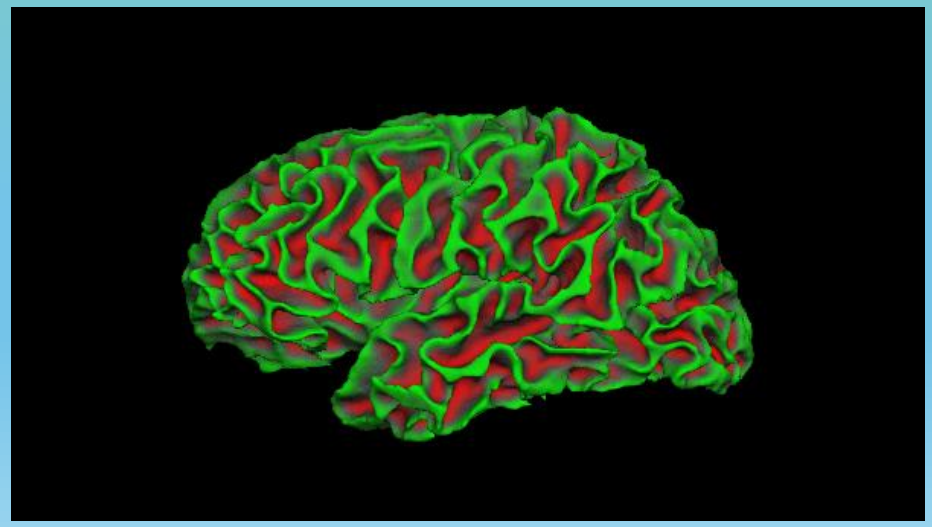
MRI Necessary



White Surface



Pial Surface



FreeSurfer

Benefits of Surface-based Analysis

- Visualization of cortical activity and effects
- Avoid Adding Partial Volume Effects (PVEs)
  - Post-reconstruction smoothing along the surface instead of 3D
  - Don’t mix tissue types at group level (vs volume with DARTEL, ANTS FSL, …)
- Only valid way to do voxel-wise analysis in cortex with Partial Volume Correction (PVC)
- Group Analysis -- better registration of cortex across subjects
- Only for cortex (GM) – subcortical still needs traditional volume-based analysis, which can still be done in PETsurfer
- PETsurfer can do full whole brain traditional volume-based analysis

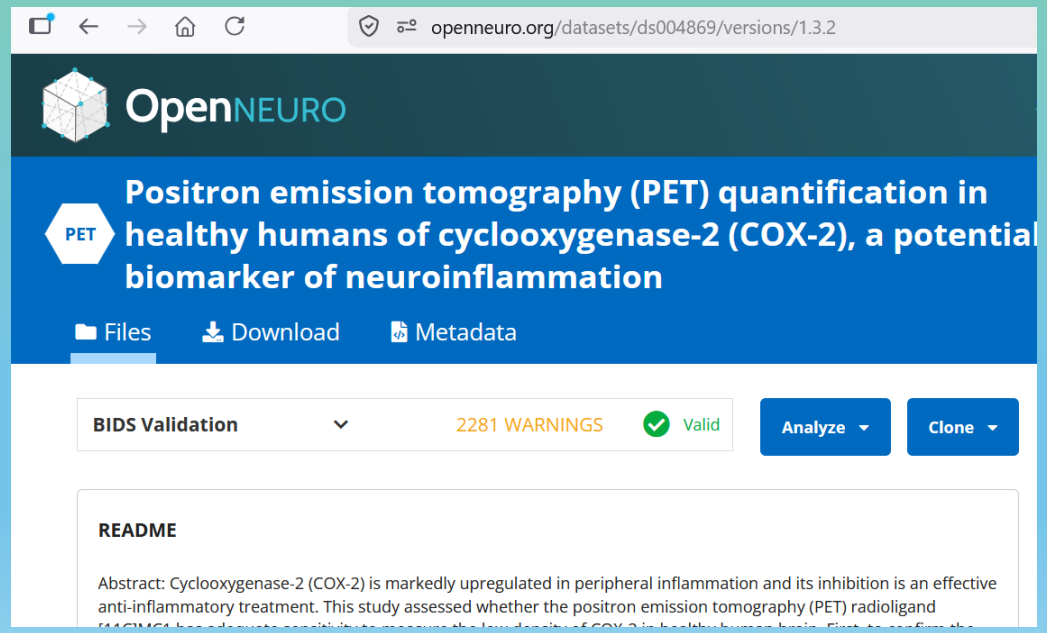
Case Study: Cyclooxygenase-2 (COX-2) [^{11}C]MC1

FEATURED CLINICAL INVESTIGATION ARTICLE

PET Quantification in Healthy Humans of Cyclooxygenase-2, a Potential Biomarker of Neuroinflammation

Xuefeng Yan[1], Martin Noergaard[1,2], Cheryl L. Morse[1], Jeih-San Liow[1], Jinsoo Hong[1], Douglas Greve[3], Sanjay Telu[1], Min-Jeong Kim[1], Jose A. Montero Santamaria[1], Anthony Galassi[1], Ningping Feng[1], Sarah K. Williams Avram[1], Ted B. Usdin[1], Shawn Wu[1], Andrea Zhang[1], Lester S. Manly[1], Madeline Jenkins[1], Maia Van Buskirk[1], Adrian Lee[1], Sami S. Zoghbi[1], Victor W. Pike[1], Paolo Zanotti-Fregonara[1], and Robert B. Innis[1]

https://openneuro.org/datasets/ds004869/versions/1.3.2



- 27 Subjects
- 10 Subjects
  - Baseline
  - Blocking
- 17 Subjects
  - Test
  - Retest

Journal of Nuclear Medicine Mar 2025, 66 (3) 398-404; DOI: 10.2967/jnumed.124.268525

Case Study: Cyclooxygenase-2 (COX-2) [^{11}C]MC1

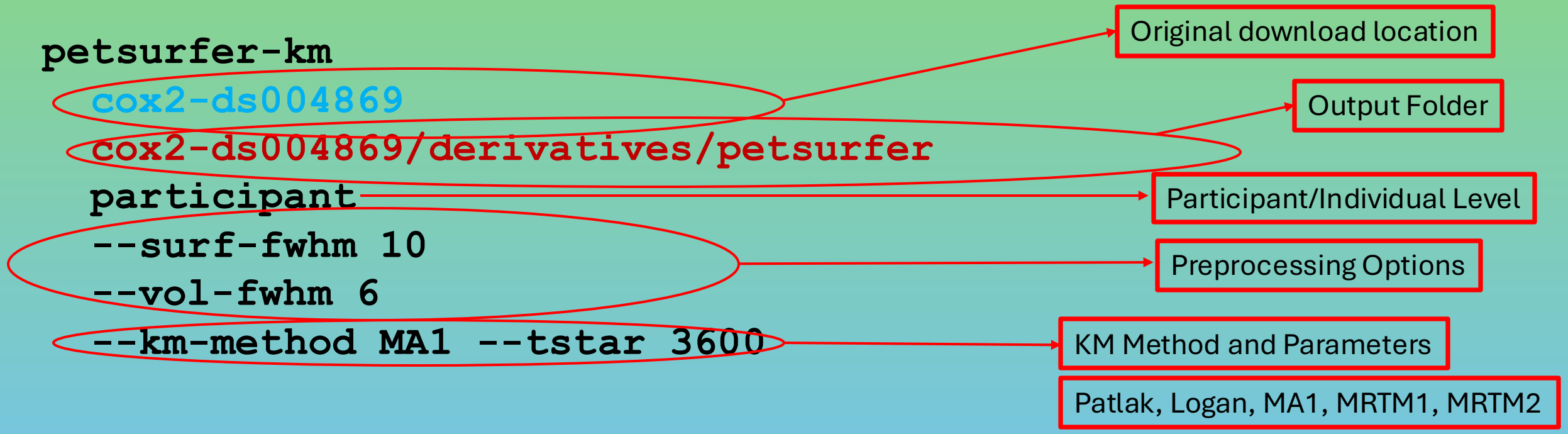
Preparation (Once)

- Download PETSurfer Container
- Download PETsurfer BIDS App
  - https://github.com/freesurfer/petsurfer-bids

- Get your data into BIDS format
- Download from OpenNeuro
  - **aws s3 sync --no-sign-request s3://openneuro.org/ds004869 cox2-ds004869**
- Run through PETprep
- Run through PETfit/BloodStream (if invasive KM)

Participant/Individual Kinetic Modeling

```
petsurfer-km
 cox2-ds004869
 cox2-ds004869/derivatives/petsurfer
  participant
  --surf-fwhm 10
  --vol-fwhm 6
  --km-method MA1 --tstar 3600
```



- This will run all sessions/time points (baseline, blocked, test, retest) of all participants
- All tracers found (Cox2 is MC1)
- Volume, surface, and ROI

Download command: **aws s3 sync --no-sign-request s3://openneuro.org/ds004869 cox2-ds004869**

Individual Kinetic Modeling: Summary Web Page

petsurfer-km Summary Parametric Maps ROI Results About sub-01_ses-baseline

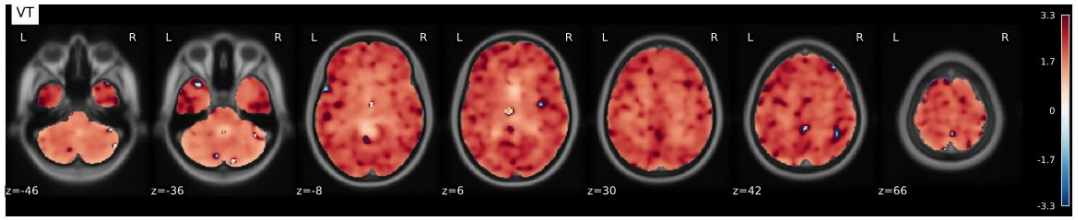
Summary

Subject	sub-01
Session	ses-baseline
Tracer	MC1
Methods	MA1
t* (seconds)	3600.0
Volumetric smoothing (FWHM)	6.0 mm
Surface smoothing (FWHM)	10.0 mm

Parametric Maps

MA1 — VT

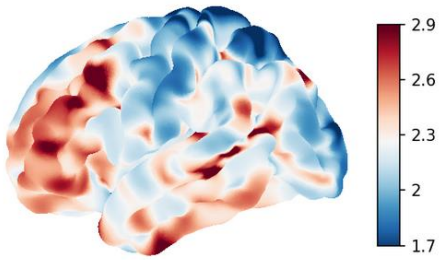
Volumetric (MNI152)



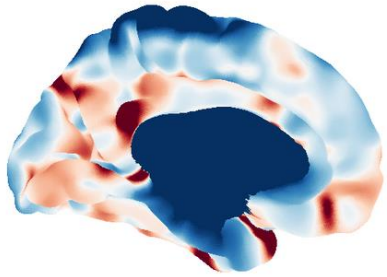
View results in freebrowse

Surface (LH, fsaverage)

VT (LH, lateral)



VT (LH, medial)



ROI Results

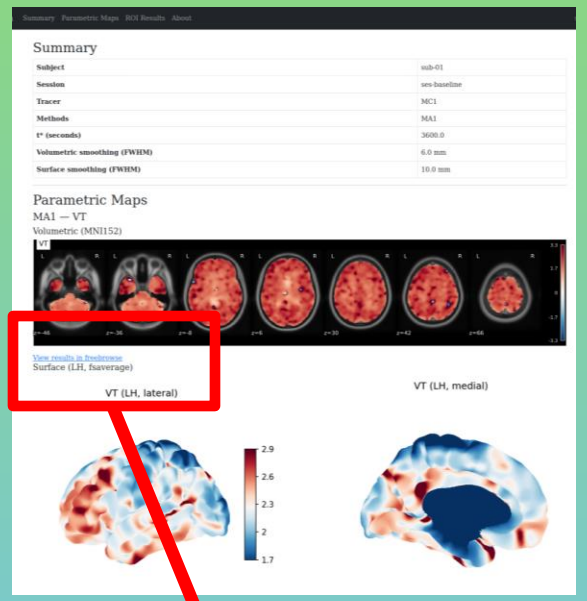
MA1 — VT

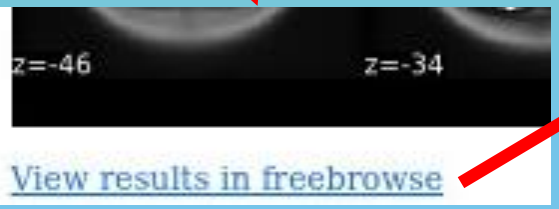
Region	VT
Left-Cerebral-White-Matter	2.05846
Left-Lateral-Ventricle	1.31822
Left-Inf-Lat-Vent	1.67997
Left-Cerebellum-White-Matter	1.68937
Left-Cerebellum-Cortex	1.65865
Left-Thalamus	1.67841
Left-Caudate	1.49408
Left-Putamen	1.97959
Left-Pallidum	1.91267
Brain-Stem	1.42120
Left-Hippocampus	1.57259
Left-Amygdala	1.73245
CSF	1.26937

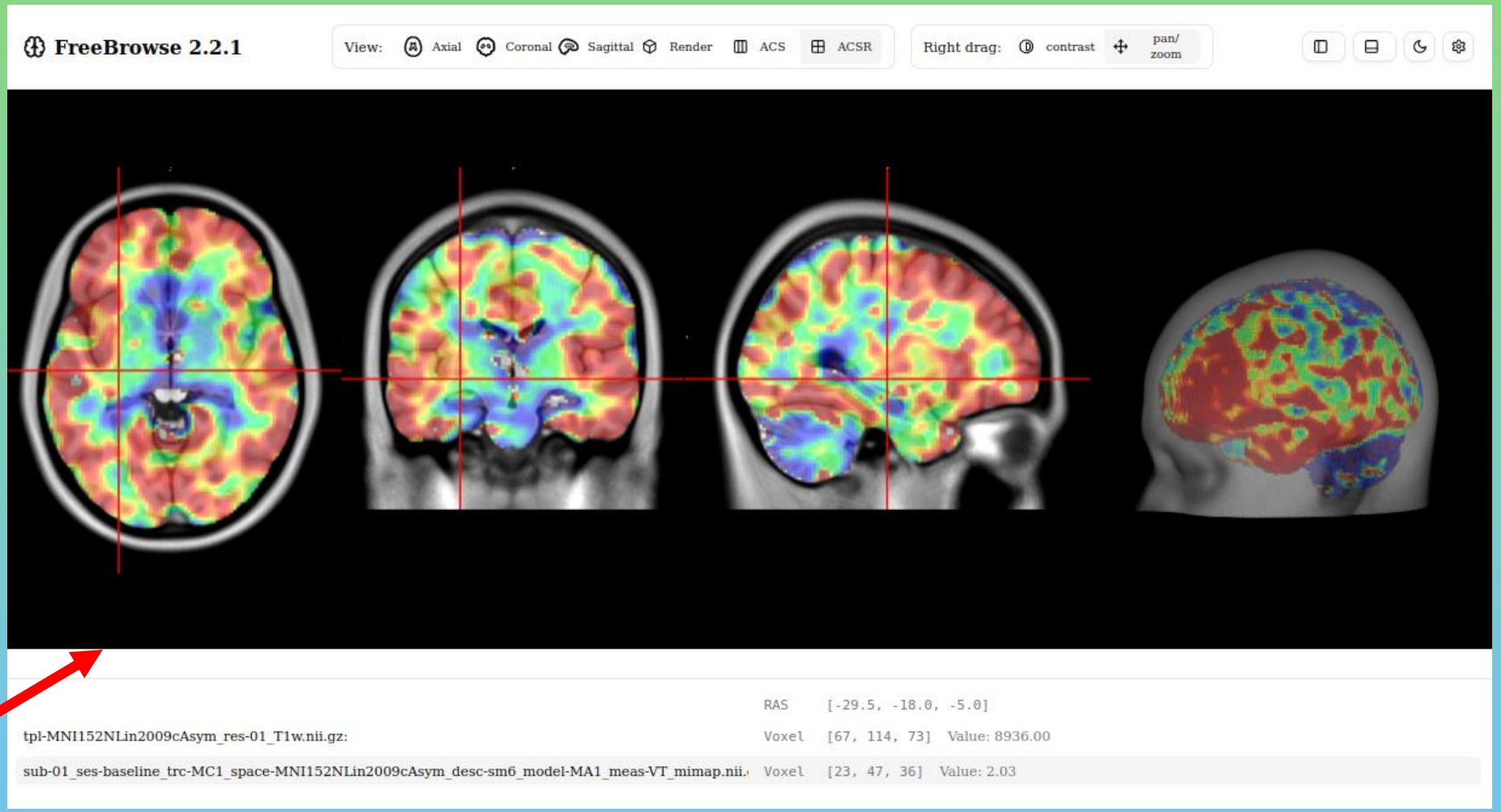
About

petsurfer-km version	0.2.3
Invocation Command	`/opt/petsurfer-bids/bin/petsurfer-km /data/input /data/output participant --participant-label 01 --vol-fwhm 6 --surf-fwhm 10 --km-method logan-ma1 --tstar 3600`
Commands Executed	1. **Extract frame times from TAC file**

Interactive with FreeBrowse/NiiView







- Nothing to install
- All data stays local
- Acknowledgement to Paul Wighton

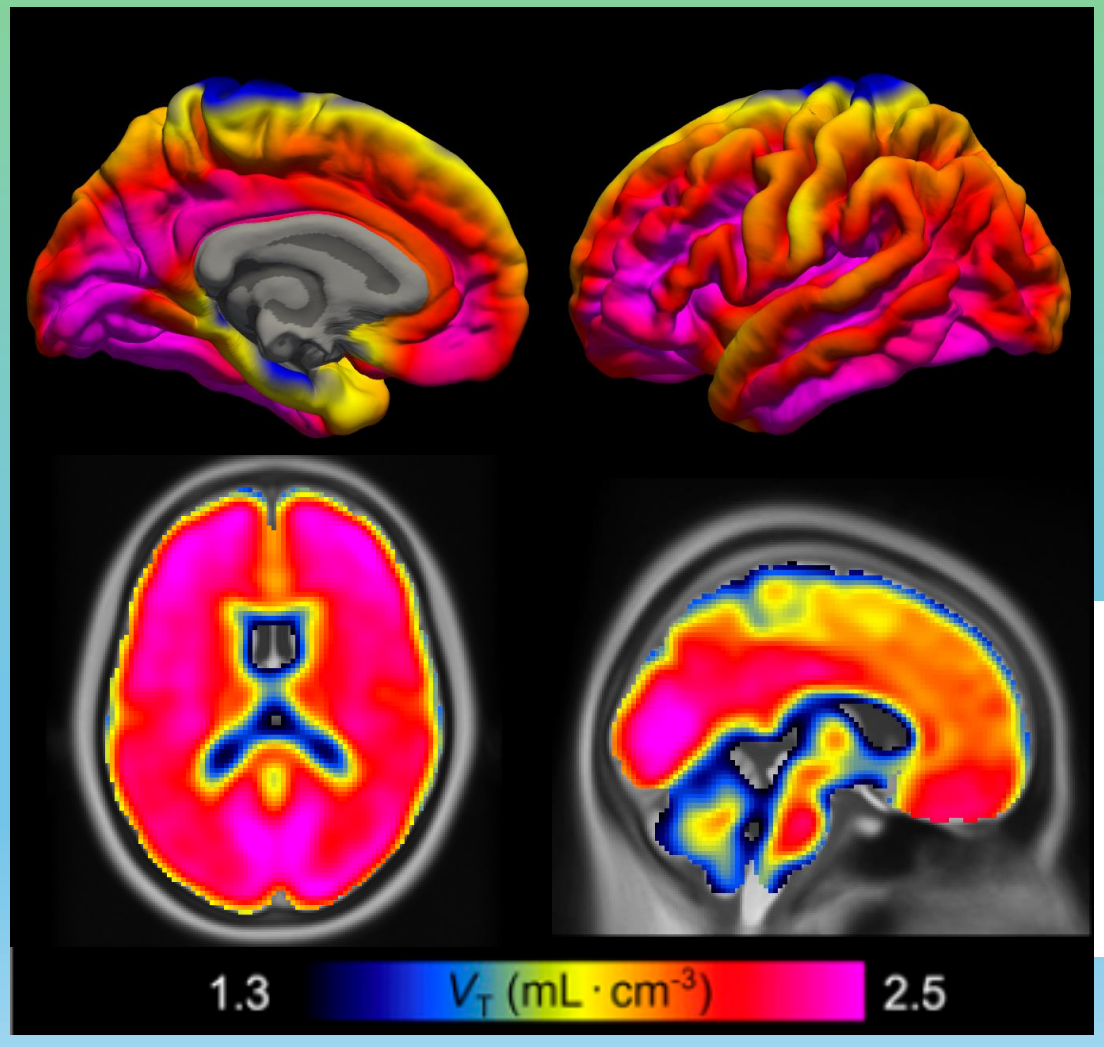
Group Analysis

```
petsurfer-km-group
  cox2-ds004869/derivatives/petsurfer
  group.analysis
  --km-method MA1
  --tracer MC1
  --spaces fsaverage-lh fsaverage-rh mni ROI
  --vol-fwhm 6
  --surf-fwhm 10
```

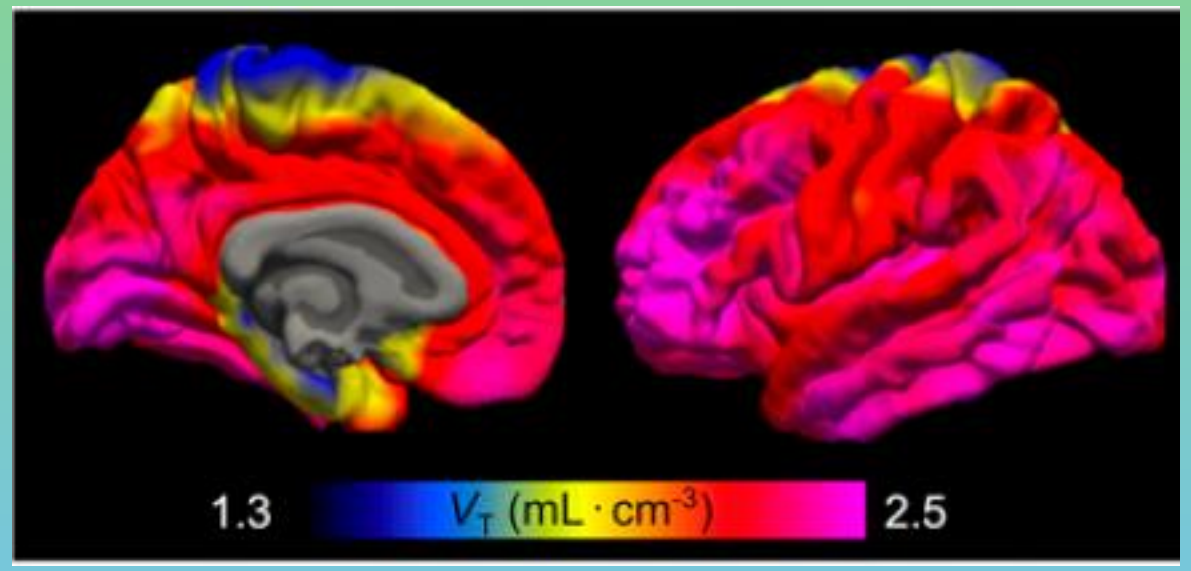
```
petsurfer-km
  cox2-ds004869
  cox2-ds004869/derivatives/petsurfer
```

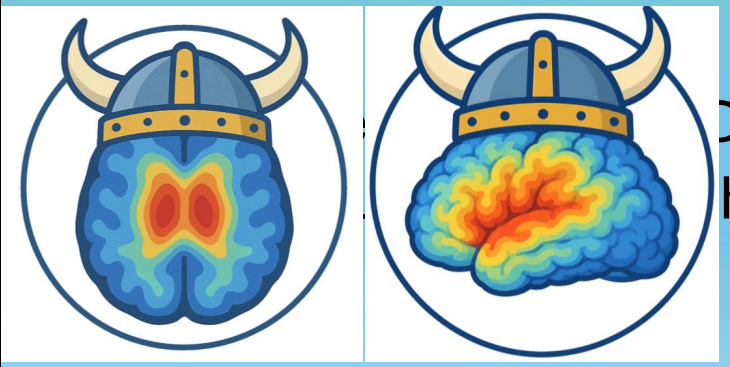
Population Mean Maps of V_T

PETsurfer, PETprep, BloodStream



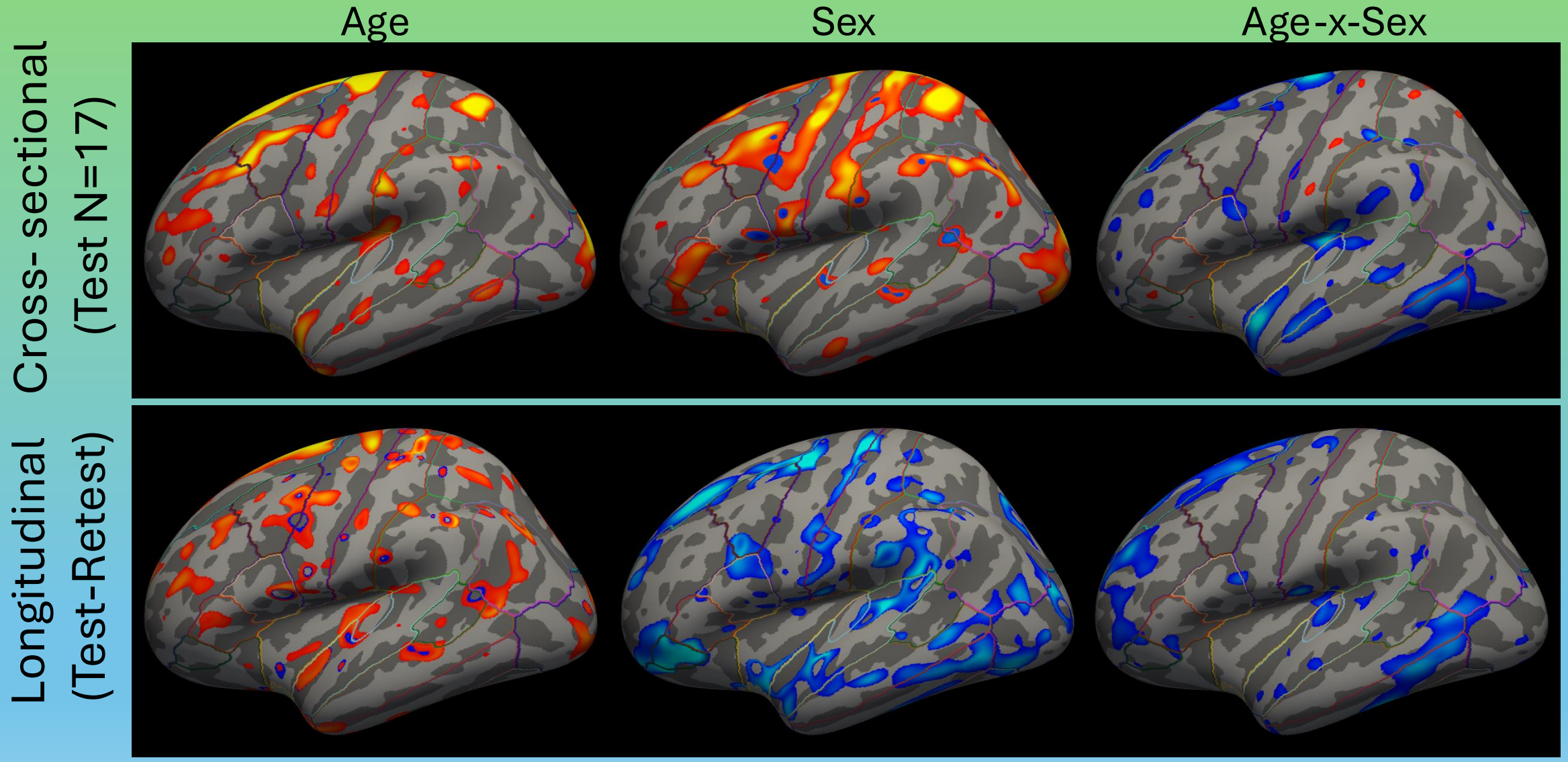
Publication Yan, et al, 2025





DS-structured
h 4 BIDS apps

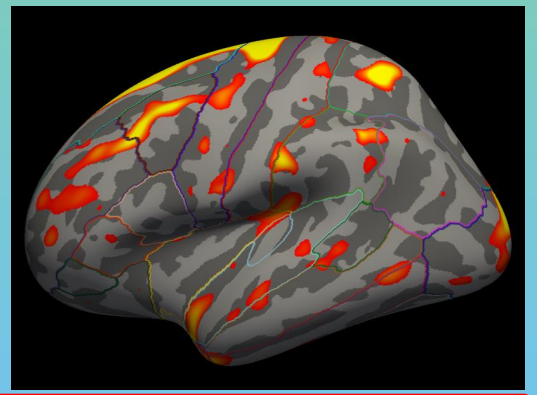
P-value Maps of Group Effects on V_T



- Note: thresholds are ridiculously low to show some effects

Group Analysis: Correction for Multiple Comparisons

```
petsurfer-km-group
   cox2-ds004869/derivatives/petsurfer
   group.analysis
   --km-method MA1
   --tracer MC1
   --spaces fsaverage-lh fsaverage-rh mni ROI
   --vol-fwhm 6
   --surf-fwhm 10
   --cmc 500 2 abs 2 0.05
```



Permutation-based Correction for Multiple Comparisons (CMC) N=500. Cluster-wise. Output table That can be inserted into Results.

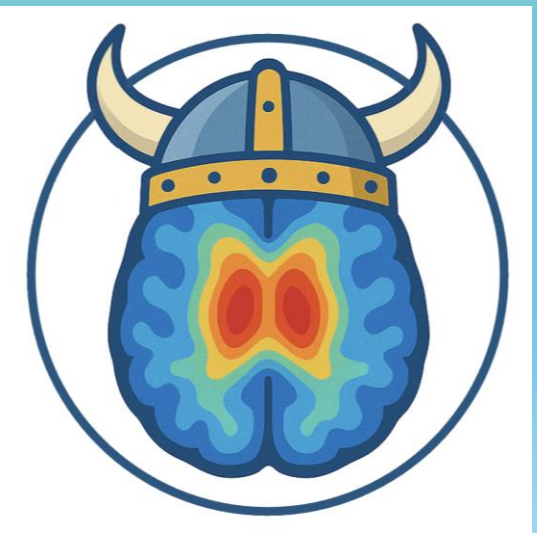
Summary: End-to-End Processing with BIDS Apps

- PETPrep – motion correction, registration, PVC, resampling to ROI, MNI152, fsaverage
- Bloodstream – creates the AIF
- PETFit – Advanced ROI analysis (TC models, etc) with web-based configuration
- PETsurfer
  - Individual KM analysis (ROI, surface, volume) with linear KM analyses (Patlak, Logan, MRTM)
    - Web Page + Interactive Map Visualization
  - Group Analysis – Population Mean Maps, Elaborate Designs, Longitudinal, Correction for Multiple Comparisons ready for publication

Thanks!

Co-Authors:
Paul Wighton
Martin Norgaard
Granville Matheson
Melanie Ganz
Cyril Pernet
Anthony Galassi
Gitte Knudsen
Robert Innis

NIH Grants
R01EB023281
R01NS083534
R01EB035560
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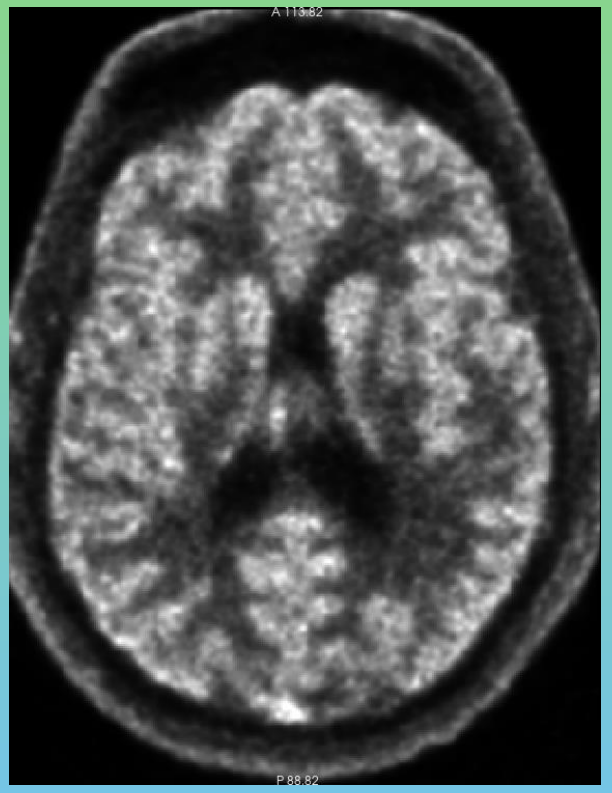


End of Presentation

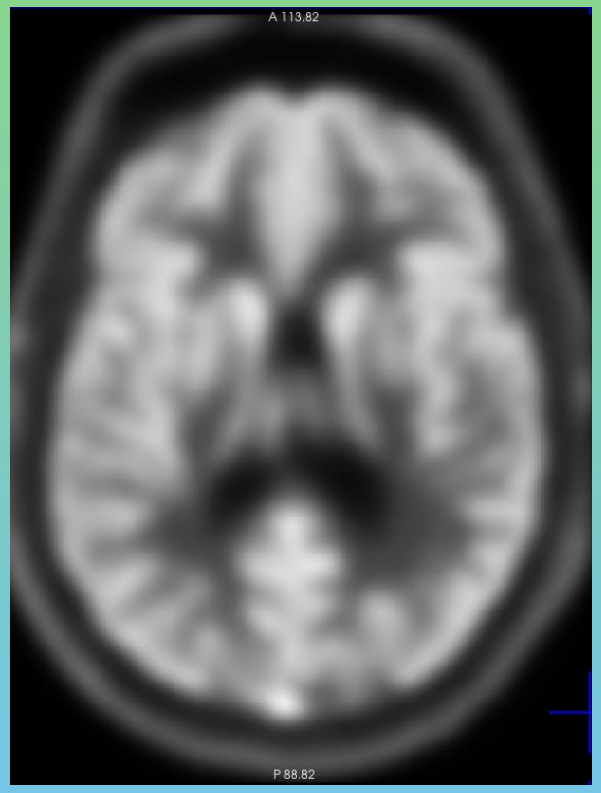
Surface-based Analysis Benefits: Visualization

Surface-based Analysis Benefits: Spatial smoothing without PVEs

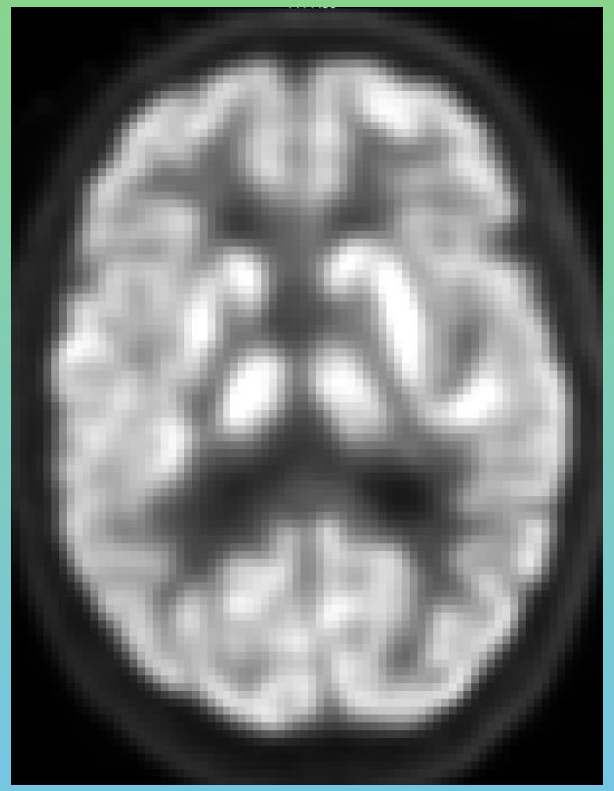
Traditional Volume-based Analysis and Smoothing



Yale NX FDG
~2025

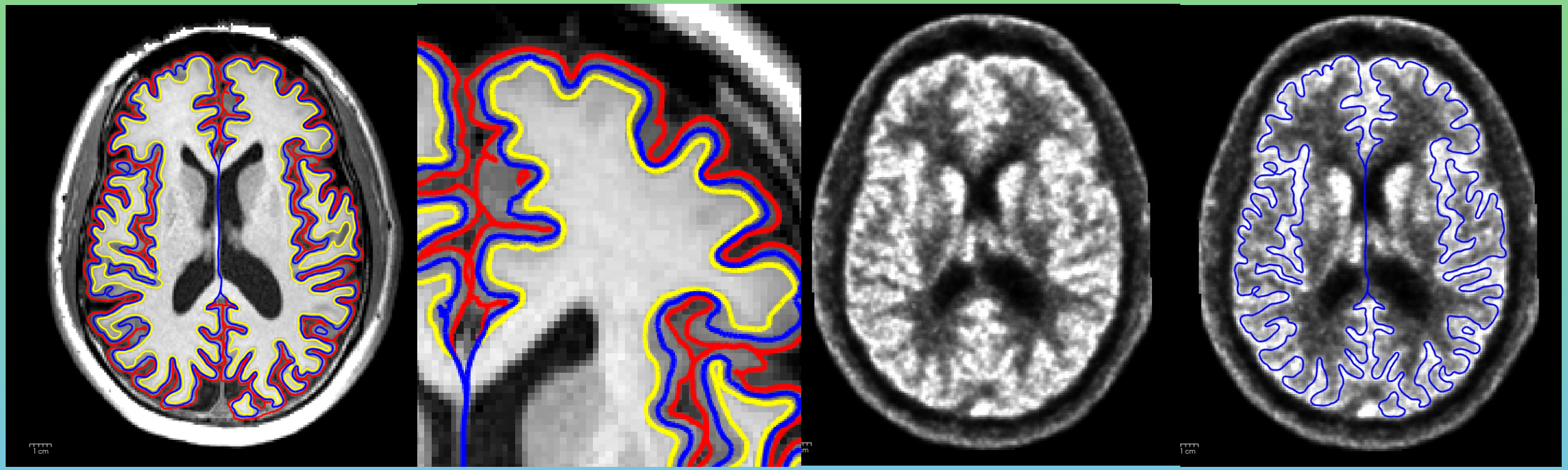


Smoothed by 6mm



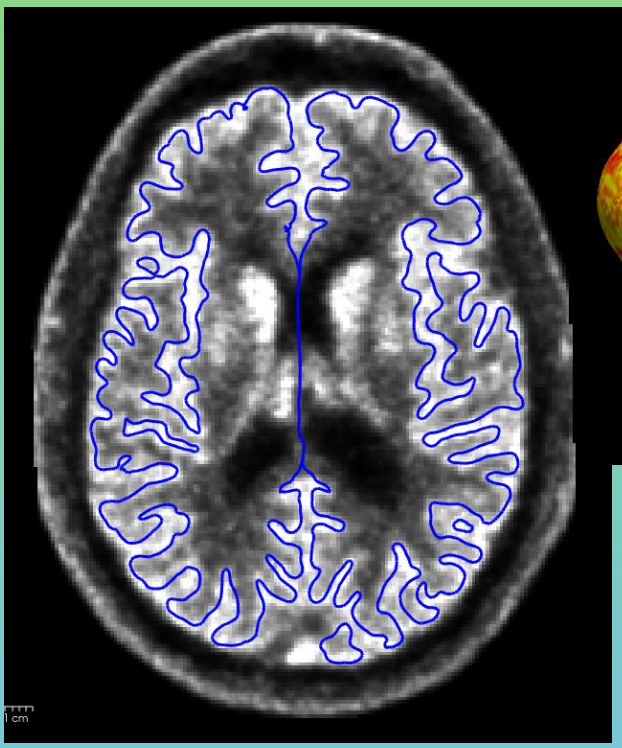
Siemens HR+
~1995

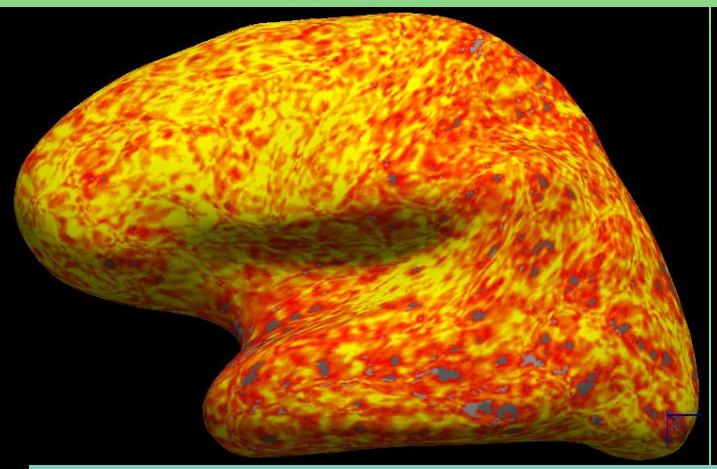
Surface-based Analysis and Smoothing

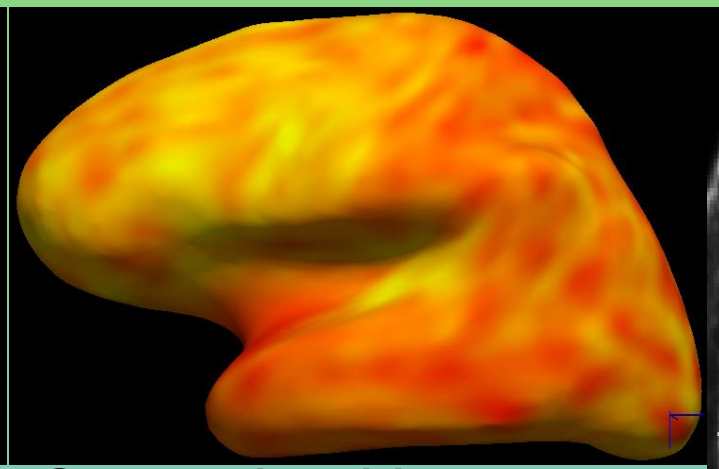


Blue Line is the middle of the cortical ribbon

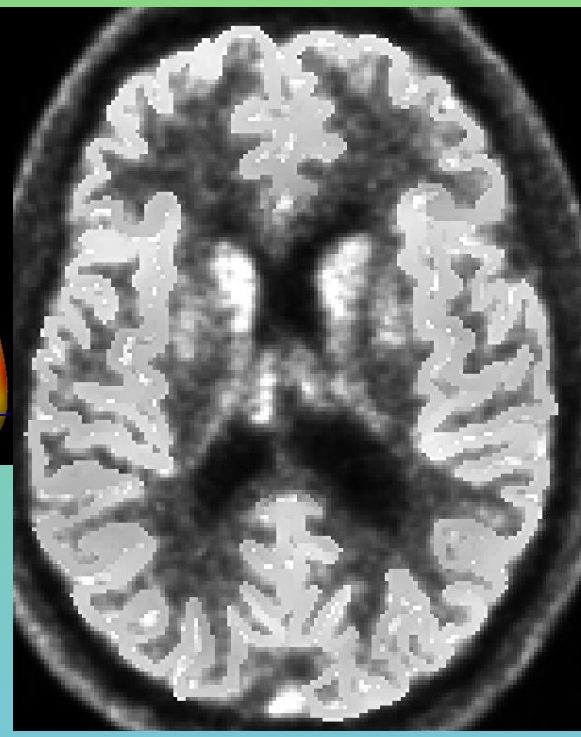
Surface Sampling and Smoothing Without PVEs

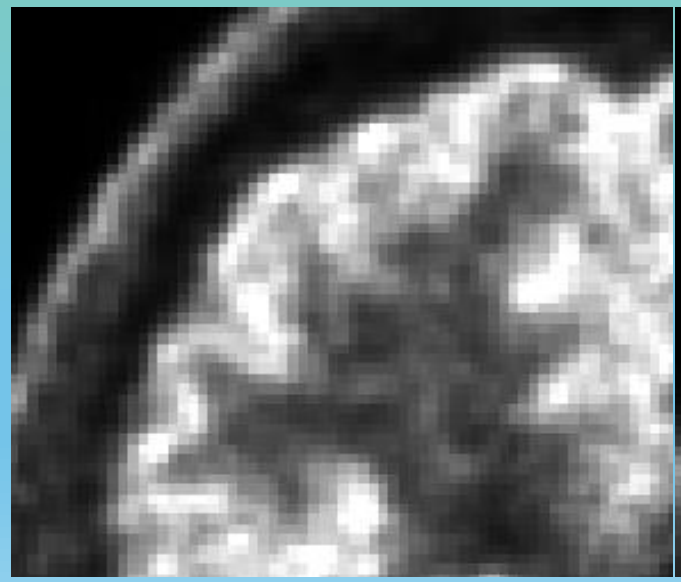


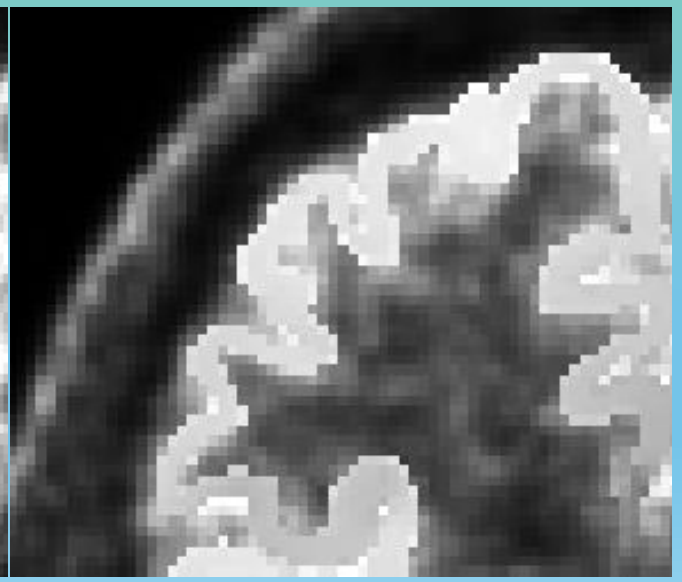




Smoothed by 10mm

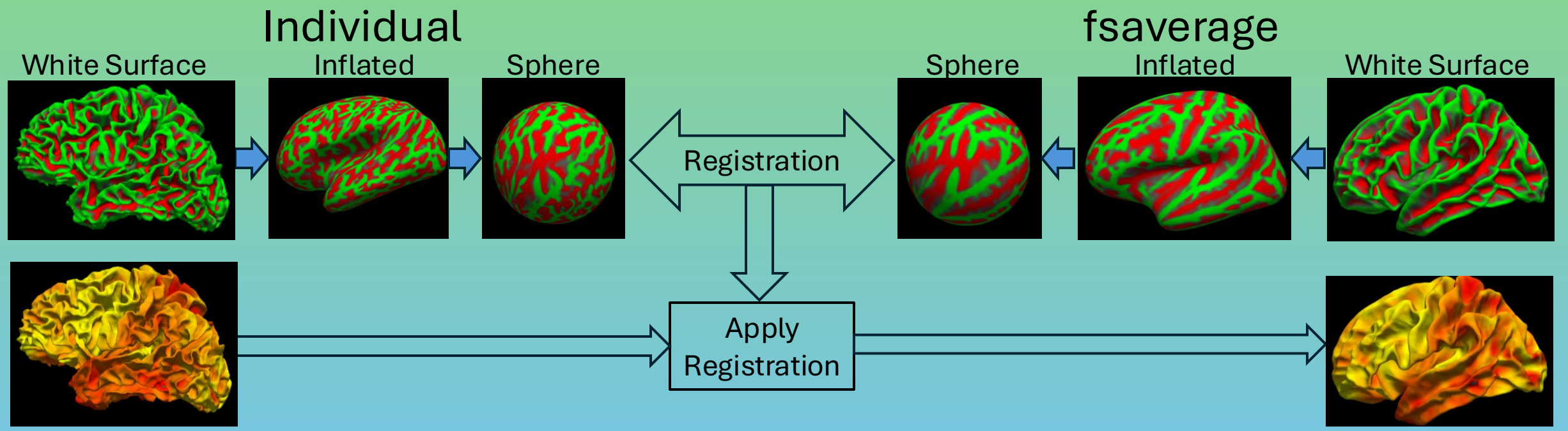






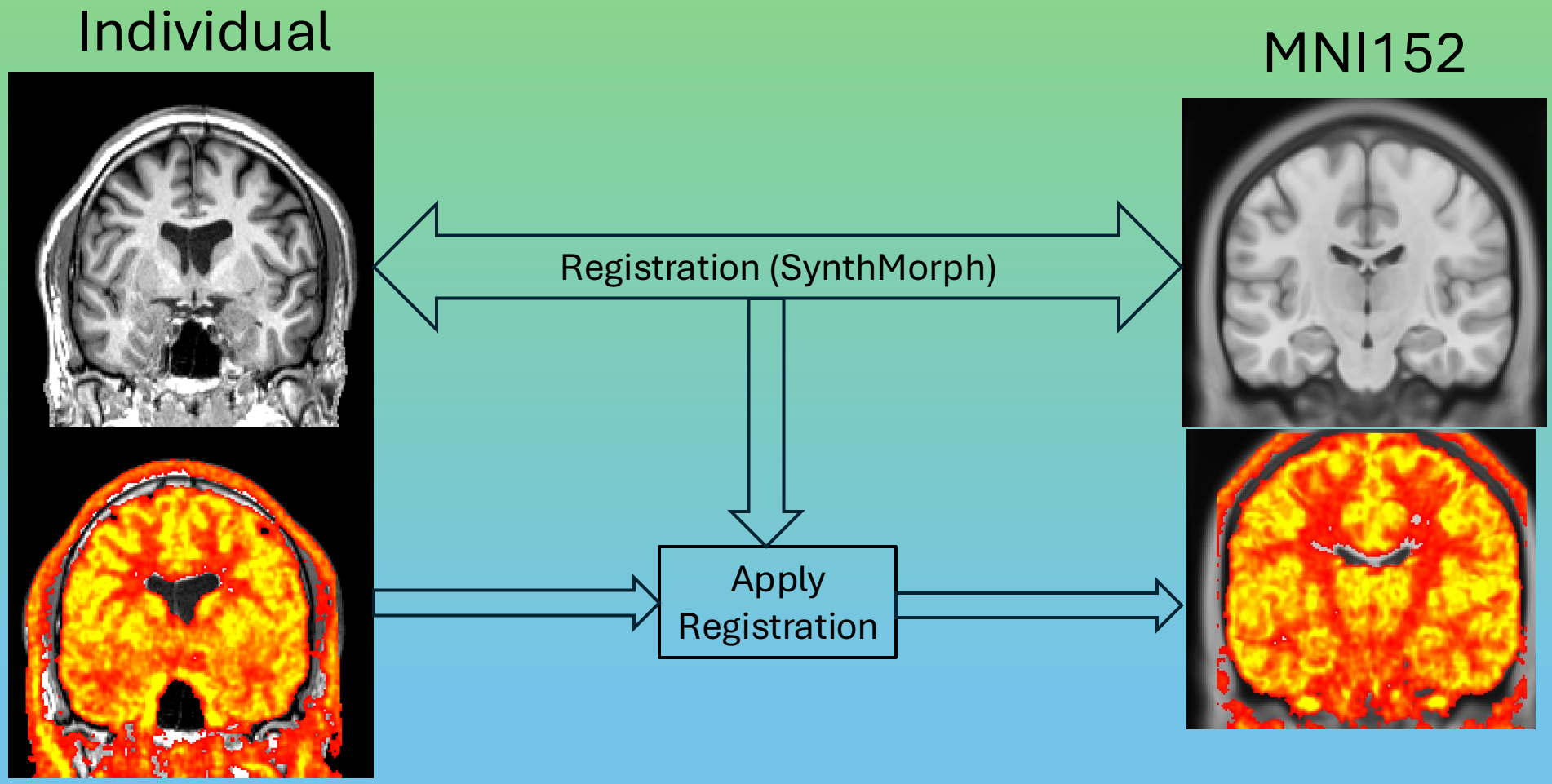
Surface-based Analysis Benefits: Inter-Subject Registration

Surface-based Inter-subject Registration



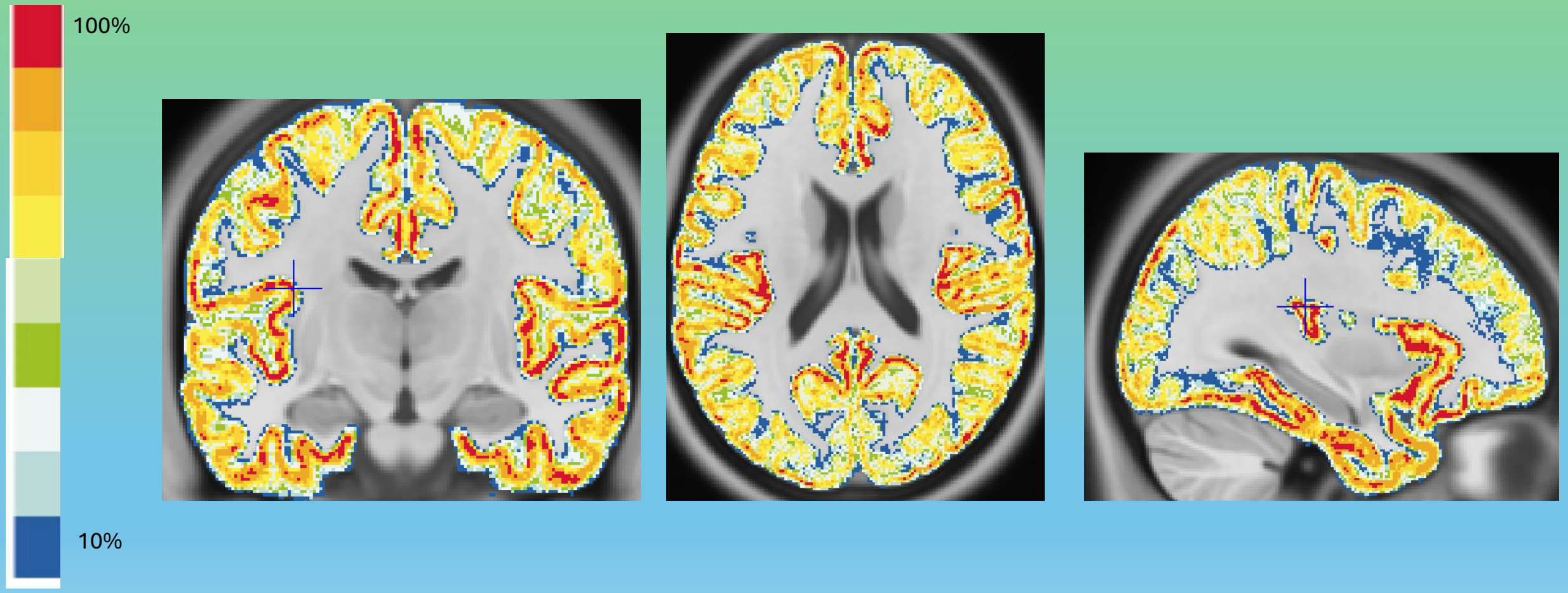
Yale NX FDG, surface-smoothed by 10mm

Volume-based Inter-subject Registration



Yale NX FDG

How well does volume-based Inter-subject registration do in cortex?

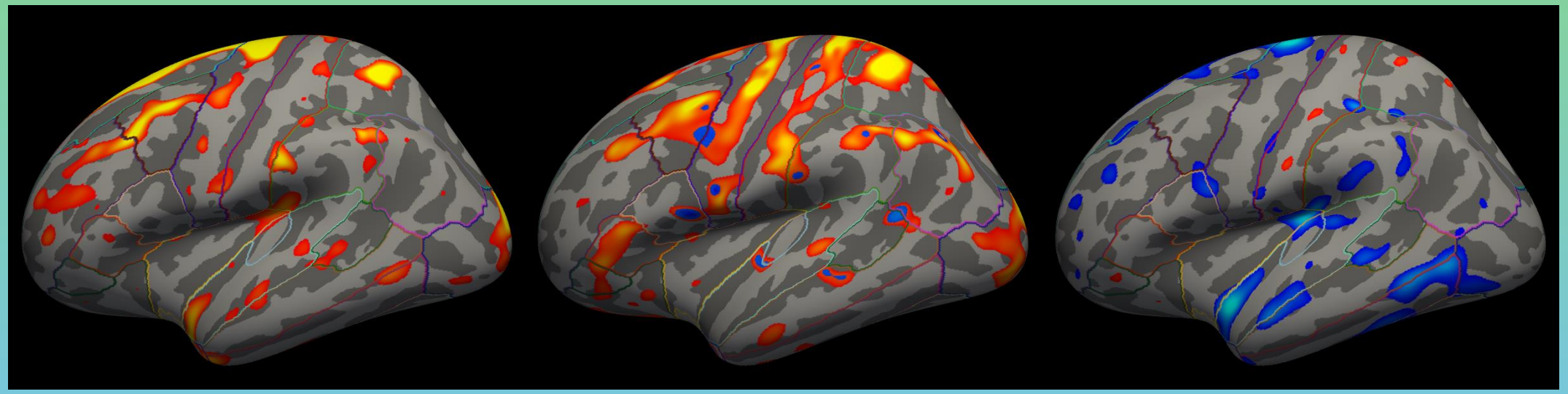


Benefits of Surface-based Analysis

- Visualization of cortical activity and effects
- Spatial smoothing without PVEs
- Only valid way to do voxel-wise analysis with PVC in cortex
- Group Analysis
  - Better registration of cortex across subjects
  - Gray matter only. Vs volume-based methods (SPM, FSL, ANTS, SynthMorph) will result in GM and WM being at the same voxel.
- Only for cortex – subcortical still needs traditional volume-based analysis, which can still be done in PETsurfer

P-value Maps of Group Effects on V_T

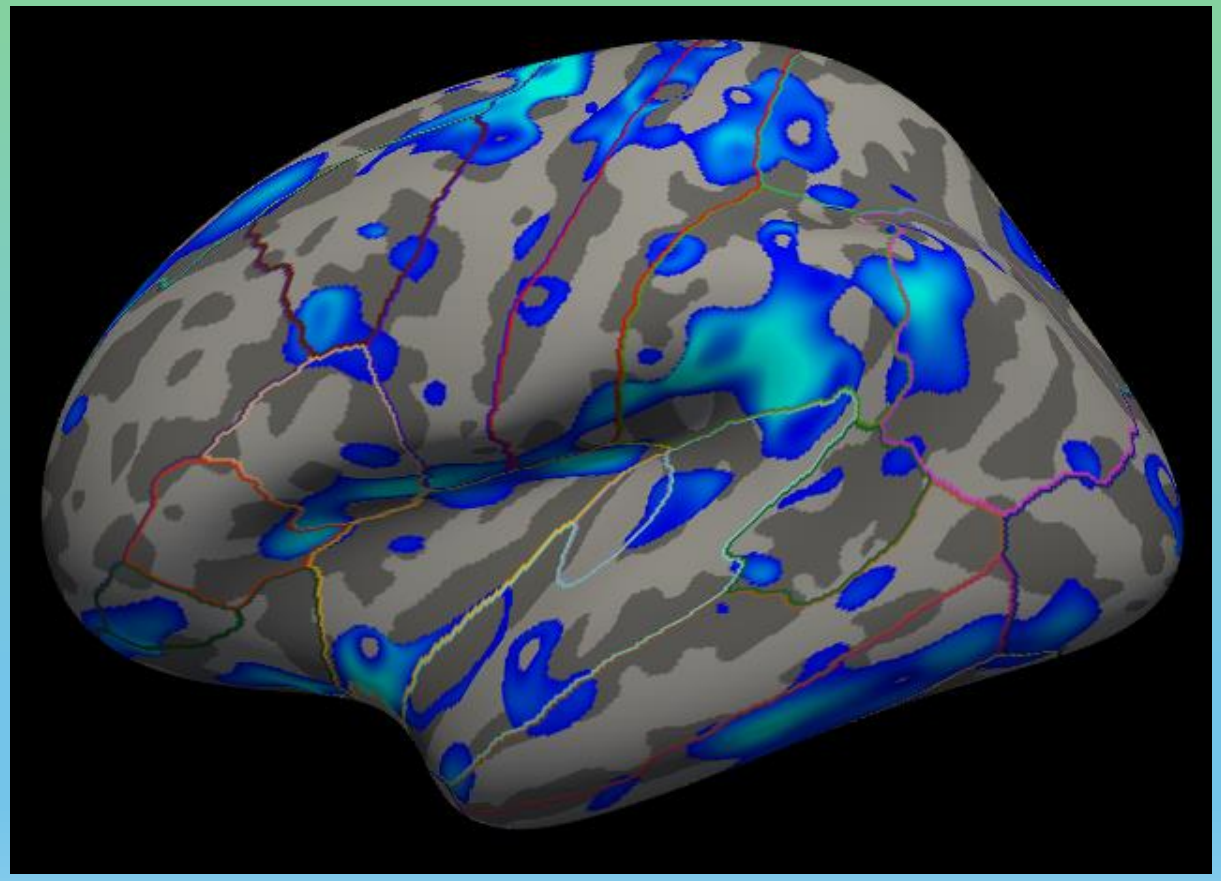
Age Sex Age-x-Sex



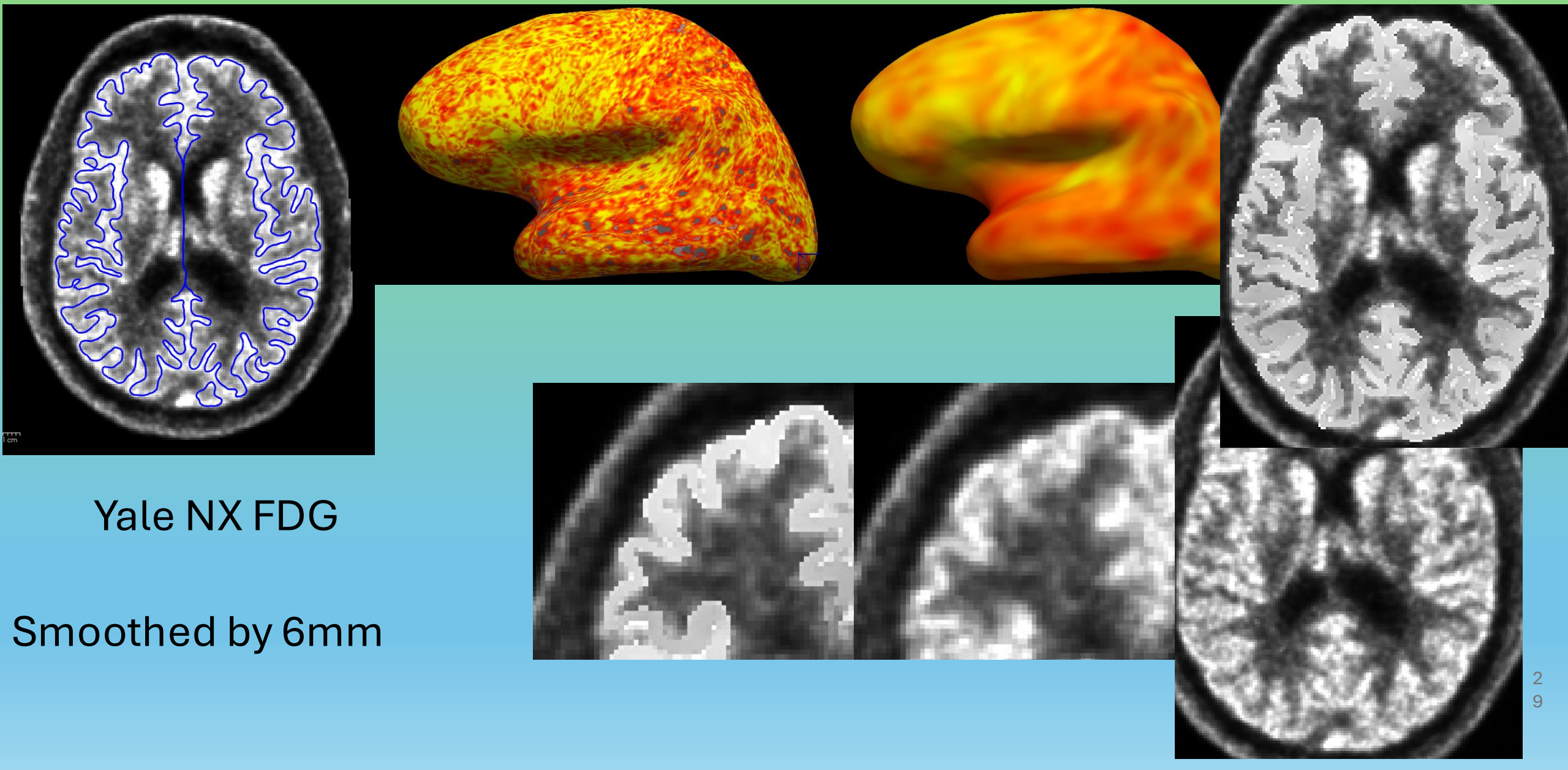
- Note: thresholds are ridiculously low to show some effects

Group Analysis of KM: Longitudinal

apptainer run



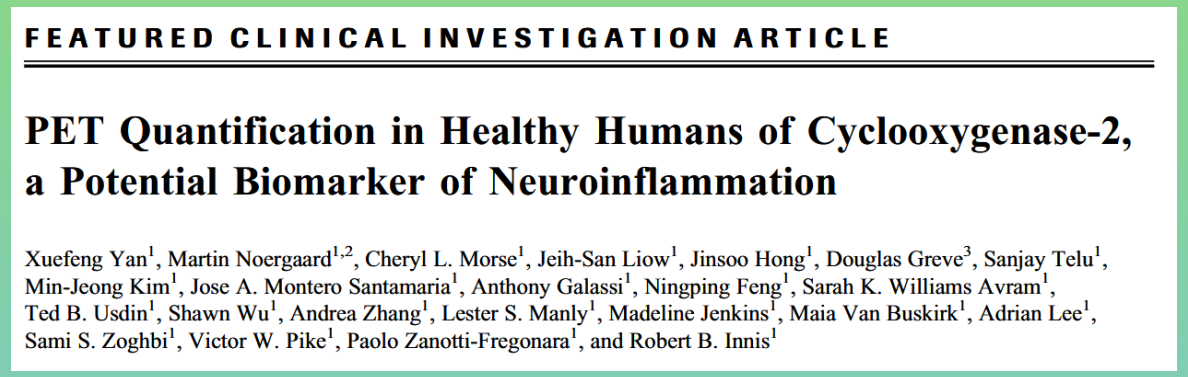
Surface-based Analysis and Smoothing



Yale NX FDG

Smoothed by 6mm

Case Study: Cyclooxygenase-2 (COX-2) [11C]MC1

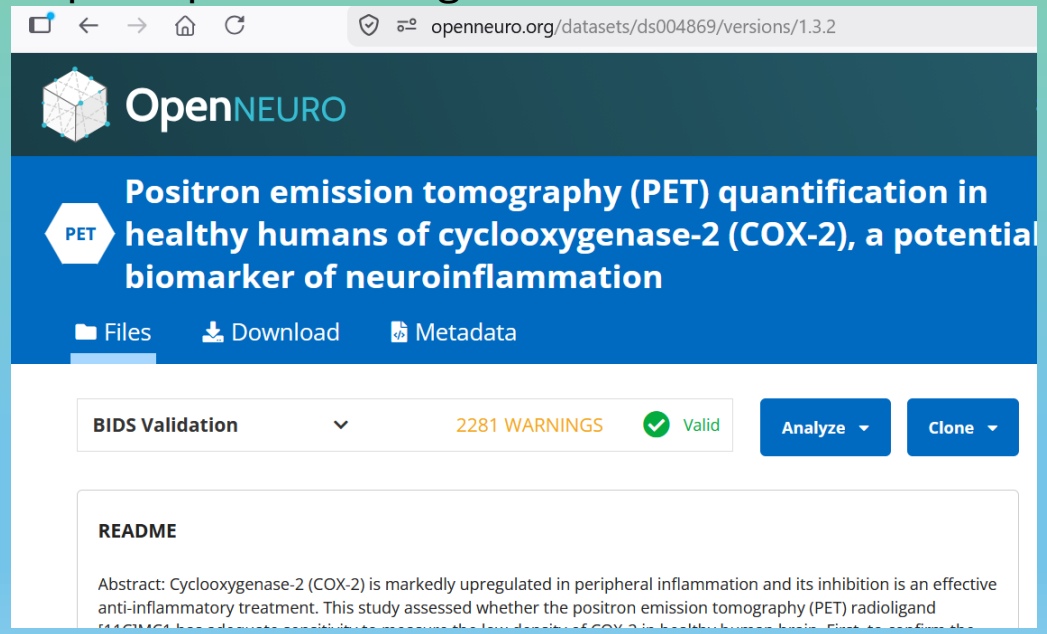


FEATURED CLINICAL INVESTIGATION ARTICLE

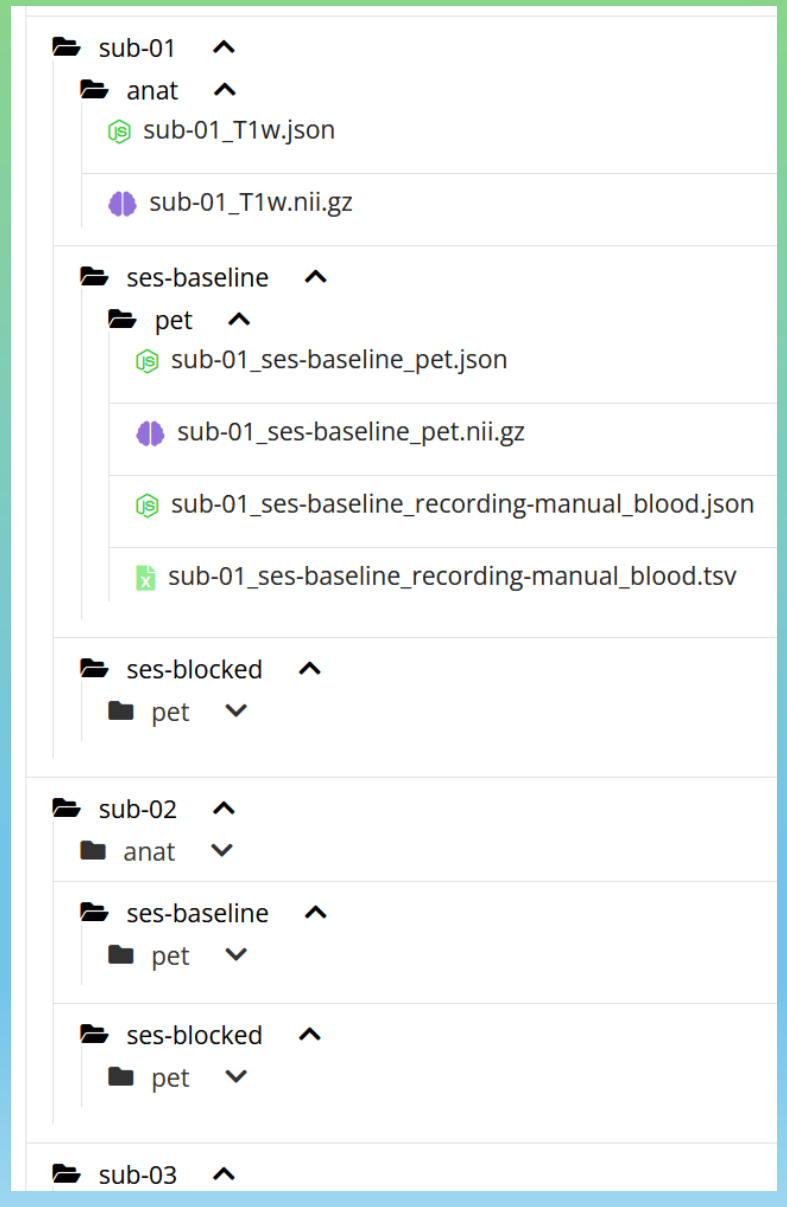
PET Quantification in Healthy Humans of Cyclooxygenase-2, a Potential Biomarker of Neuroinflammation

Xuefeng Yan[1], Martin Noergaard[1,2], Cheryl L. Morse[1], Jeih-San Liow[1], Jinsoo Hong[1], Douglas Greve[3], Sanjay Telu[1], Min-Jeong Kim[1], Jose A. Montero Santamaria[1], Anthony Galassi[1], Ningping Feng[1], Sarah K. Williams Avram[1], Ted B. Usdin[1], Shawn Wu[1], Andrea Zhang[1], Lester S. Manly[1], Madeline Jenkins[1], Maia Van Buskirk[1], Adrian Lee[1], Sami S. Zoghbi[1], Victor W. Pike[1], Paolo Zanotti-Fregonara[1], and Robert B. Innis[1]

https://openneuro.org/datasets/ds004869/versions/1.3.2



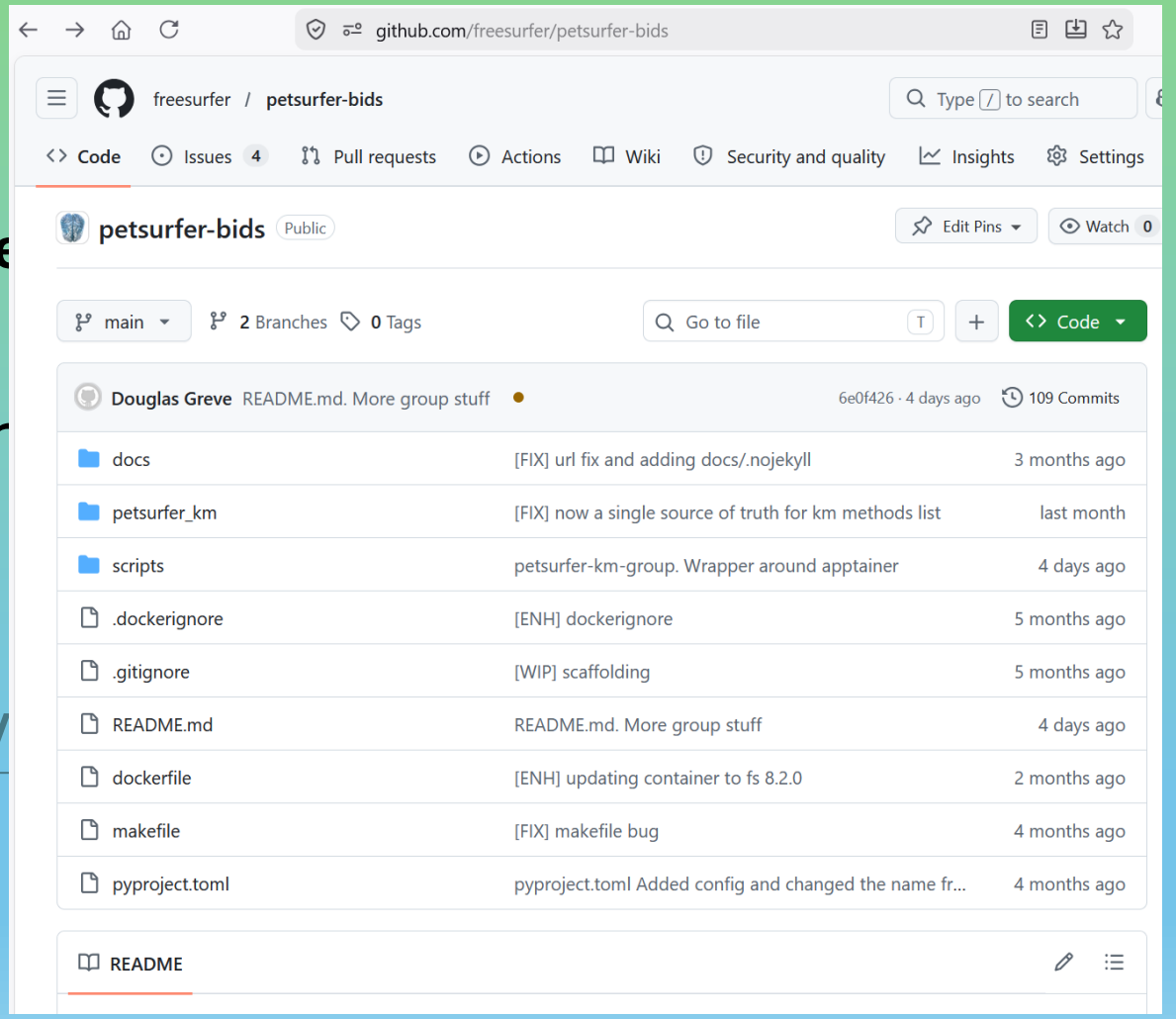
Journal of Nuclear Medicine Mar 2025, 66 (3) 398-404; DOI: 10.2967/jnumed.124.268525



Case Study: Cyclooxygenase-2 (COX-2) [11C]MC1

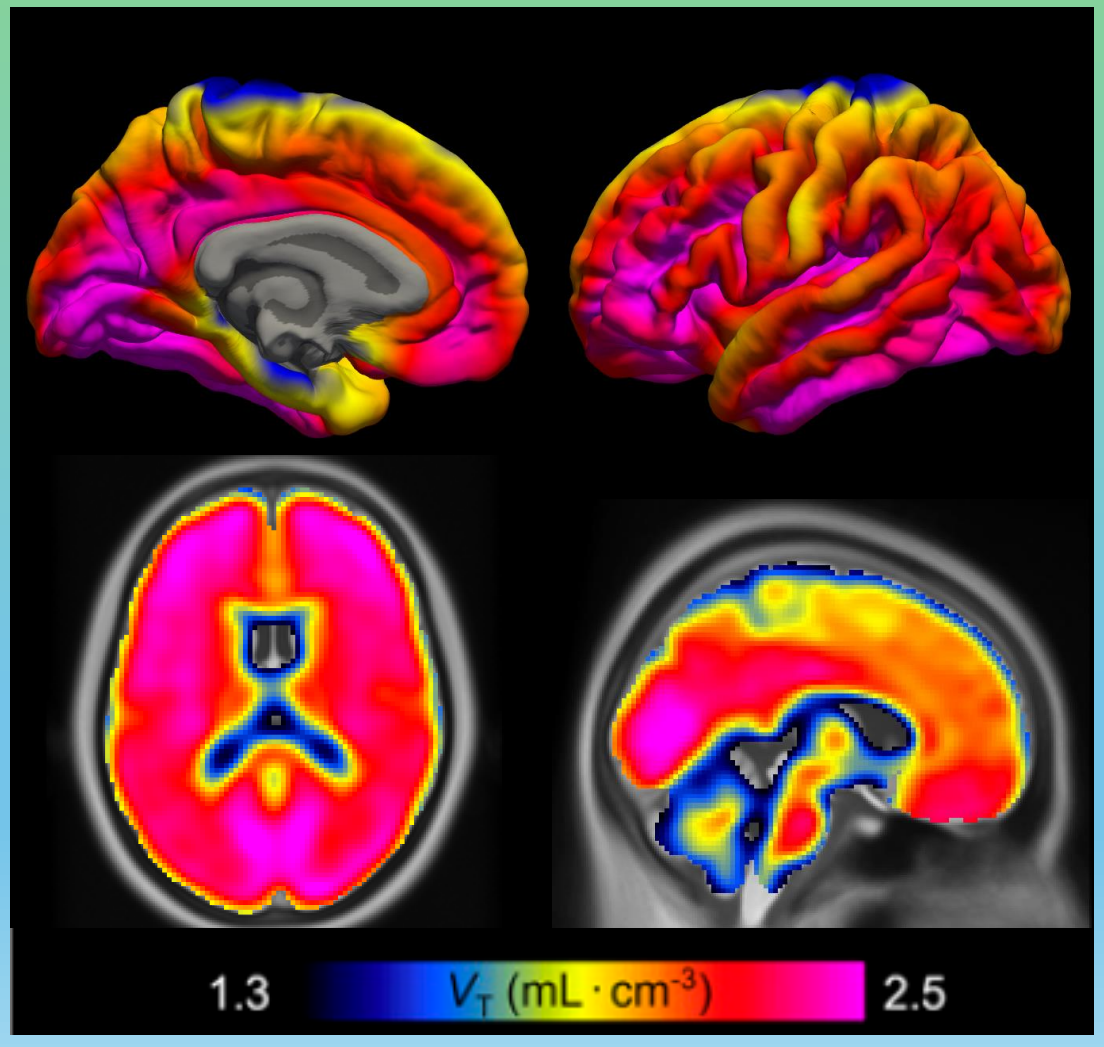
Preparation

- Download from OpenNeuro
  - **aws s3 sync --no-sign-request s3://ope**
- Run through PETPrep
- Run through KinFitr/BloodStream
- Container
  - PETsurfer BIDS App
  - https://github.com/freesurfer/

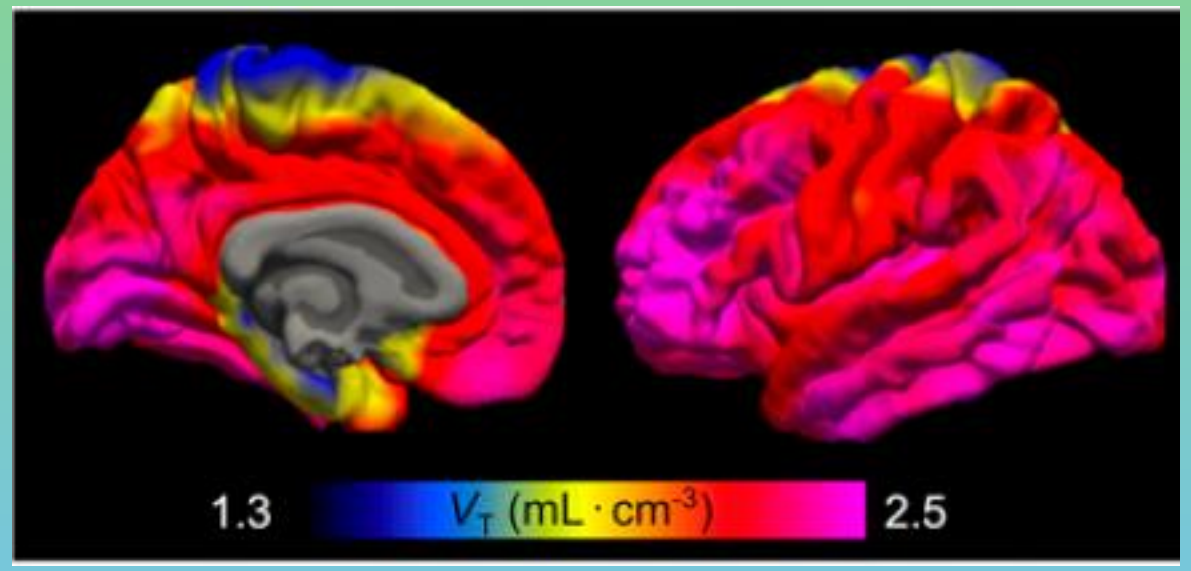


Population Mean Maps of V_T

PETsurfer, PETprep, BloodStream



Publication Yan, et al, 2025



Go all the way from BIDS-structured data to publication with 4 BIDS apps

Benefits of Surface-based Analysis

- Visualization of cortical activity and effects
- Spatial smoothing without adding Partial Volume Effects (PVEs)
- Only valid way to do voxel-wise analysis in cortex with Partial Volume Correction (PVC)
- Group Analysis
  - Better registration of cortex across subjects
  - Gray matter only. No PVEs at the group level. Vs volume-based methods (SPM, FSL, ANTS, SynthMorph) will result in GM and WM being mixed.
- Only for cortex – subcortical still needs traditional volume-based analysis, which can still be done in PETsurfer
- PETsurfer can do full whole brain traditional volume-based analysis